MID SEMESTER-2026A
STATISTICAL COMPUTING

Name: Roll Number:

TIME: 2 HRS
MAXIMUM MARKS: 40

This paper has seven questions. Questions 1-3 are subjective questions and you have to show all your work to get full marks. To write an algorithm you have to write as Step-1, Step-2, etc, otherwise you will not get any credit. Moreover, it is assumed that you only know how to generate samples from a uniform random variable. If you do these problems at the beginning of your answer script sequentially, you will get extra 2 points. Questions 4-7 are short answer type questions and there is no partial marking. You have to just write down the answers in this question paper in the designated places. No work needs to be shown. We will not check any rough work. Do not do any rough work in the question paper, you will loose two points. Write your name and roll numbers at the top, otherwise two marks will be deducted. Submit this question paper along with the answer booklet. We will be using the following notations: probability density function (PDF), $X \sim \text{Gamma}(\alpha, \lambda)$, for $\alpha > 0$, $\lambda > 0$, means X has PDF $f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$, for $x > 0$. $X \sim \text{GE}(\alpha)$ for $\alpha > 0$, means X has PDF $f(x) = \alpha e^{-x}(1-e^{-x})^{\alpha-1}$, for $x > 0$.

1. The PDF $f(x)$ of X for $5 \le x \le 10$, is $f(x) = cx^5 e^{-x}$, and 0, otherwise. Here c is the normalizing constant. (a) Provide an algorithm to generate random sample from X without explicitly calculating c. (b) Using simulation technique find the constant c. (5+3=8)

2. (a) Let $\{N_1(t); t \ge 0\}$ be a Poisson process with parameter 2. Provide an algorithm to generate a sample path of the above process $\{N_1(t); 0 \le t \le 10\}$ and draw the path. (b) Let $\{N_1(t); t \ge 0\}$ and $\{N_2(t); t \ge 0\}$ be two independent Poisson processes with parameters 2 and 4, respectively. Let $N(t) = N_1(t)+, N_2(t)$. Provide an algorithm to generate a sample path of the above process $\{N(t); 0 \le t \le 10\}$. [4+4 = 8]

3. The PDF $f(x)$ of X, for $x > 0$, is as follows:

$$f(x) = p_1\alpha_1 x^{\alpha_1 - 1} e^{-x^{\alpha_1}} + p_2\alpha_2 x^{\alpha_2 - 1} e^{-x^{\alpha_2}} + p_3\alpha_3 x^{\alpha_3 - 1} e^{-x^{\alpha_3}}.$$

(a) If $\alpha_1 = 0.5, \alpha_2 = 0.4, \alpha_3 = 0.7, p_1 = 0.25, p_2 = 0.25, p_3 = 0.50$. Provide an algorithm to generate random sample from X. (b) If $\alpha_1 = 0.5, \alpha_2 = 0.4, \alpha_3 = 0.7, p_1 = 0.75, p_2 = -0.5, p_3 = 0.75$. Provide an algorithm to generate random sample from X. [3+5=8]

4. Let $U \sim \text{GE}(2)$, $V \sim \text{GE}(3)$ and $W \sim \text{GE}(4)$ be three independent random variables and $X = \max\{U, W\}$ and $Y = \max\{V, W\}$. If S is a square with the four corner points, $\{(4,1), (5,1), (5,2), (4,2)\}$, and $A = P((X,Y) \in S)$, then write A as a function of a, b, c, d, where $a = 1-e^{-1}, b = 1-e^{-2}, c = 1-e^{-4}, d = 1-e^{-5}$ [4]

Answer $A =$ $d^2b^7 - d^2a^7 - c^2b^7 + c^2a^7$
$= (b^7 - a^7)(d^2 - c^2)$

5. Suppose we want to generate random sample from a $N(0,1)$ random variable, which has the probability density function $f(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$. We want to use acceptance-rejection method with the dominating function

$$g(x) = \frac{2}{\pi(4+x^2)}.$$

If c is such

$$f(x) \le cg(x); \quad \forall\ x.$$

Then the minimum values of c is [4]

Answer $\sqrt{2\pi}$

6. Suppose we want to generate random sample from a gamma distribution with the following probability density function $f(x) = \frac{1}{2}x^2e^{-x}$ for $x > 0$; based on acceptance-rejection method with the following dominating exponential PDF $g(x) = \lambda e^{-\lambda x}$. The set of all $\lambda > 0$ which can be used as the dominating function is [4]

Answer $\lambda \in (0,1)$

7. Suppose $X \sim$ Gamma(4, 4). We want to approximate estimate $\theta = E(X^{-2})$ using importance sampling technique based on the importance function $4e^{-4x}$ for $x > 0$. Suppose $Y_1, \ldots, Y_N$ are generated samples from the PDF $4e^{-4x}$ for $x > 0$. Let $\widehat{\theta}$ denote the estimate of θ based on $\{Y_1, \ldots, Y_N\}$. Then [1+3=4]

Answer:

(a) $\widehat{\theta} = \frac{1}{N}\frac{4^3}{\Gamma 4}\sum_{i=1}^{N} Y_i$

(b) $\lim_{N\to\infty} N\ Var(\widehat{\theta}) = \frac{4^4}{(\Gamma 4)^2}$

Wishing you all the best